PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	The impact of COVID-19 Pandemic on Hemorrhagic Stroke Admissions: A Systematic Review and Meta-analysis
AUTHORS	You, Yijie; Niu, Yunlian; Sun, Fengbing; Zhang, Jian; Huang, Sheng; Ding, Peiyuan; Wang, Xuhui

VERSION 1 – REVIEW

REVIEWER	Caminiti, Caterina University Hospital of Parma, Research and Innovation Unit
REVIEW RETURNED	16-Apr-2021

GENERAL COMMENTS	This review investigates hospital admissions for hemorrhagic stroke (HS) during vs before the COVID-19 pandemic, but also reports data on overall strokes and uses them to make inferences on reasons for the change in admission rates. It is certainly a very current and interesting topic. The main problem I see with this work is a possible study selection bias, due to the fact that the authors only considered studies which reported data on HS separately. This likely lead to the exclusion of relevant publications on overall strokes, thus undermining the analysis and consequent comparisons between different stroke types. Therefore, either a different search strategy is used, or results on overall strokes should be eliminated. The discussion on differences between HS overall stroke could be made using data from published reviews as a comparison. Another important issue concerns criteria for study selection. At the end of the rationale it is stated that retrospective observational studies were reviewed. However, in the Methods section, study design is not reported as a study selection criteria. Actually, Table 1 on study characteristics does feature 1 study, by Sacco et al, defined as prospective and retrospective. TITLE: it should clearly indicate that hospitalizations are investigated ABSTRACT: the objective is not clearly stated: again, hospitalizations are the object of investigation. INTRODUCTION: as required in the PRISMA guidelines, the rationale for the review should be described in the context of what is already known (e.g. cite existing reviews or indicate that none is present), and specify the gap that this work intends to fill. METHODS: the search strategy appears very generic and not accurately described, thus it is not replicable. I suggest providing the full strategy as a supplementary file, as indicated by the PRISMA guidelines. RESULTS AND DISCUSSION: please see my introductory comments on stroke types and study selection bias. Also, at the beginning of the discussion, it should be clarified that the focus of the study is admissions, and not patients.

REVIEWER	Salam, Abdul

	Hamad General Hospital
REVIEW RETURNED	22-Apr-2021

GENERAL COMMENTS	Reviewer Comments to Author Thank you for this interesting systematic review and meta-analysis to study the Impact of the Coronavirus Disease Pandemic on the hemorrhagic stroke: A Systematic Review and Meta-analysis. Below are my suggestions and remarks. Abstract: Methods: The authors need to specify the statistical analysis methods used in the manuscript. Methods: The authors used EMBASE, PubMed, and Web of Science electronic databases. Why authors didn't use other sources such as MEDLINE—In Process, Elsevier, Google Scholar, and Cochrane Database of Systematic Reviews (CDSR). Page 7 line 12: "A fixed-effect model was used when statistically indicated heterogeneity was found (P>.1, I2<50%)" Double check this line I believe you mean A fixed-effect model was used where there was no heterogeneity (P>.1, I2<50%). The authors wrote that Sensitivity analysis was used to exclude the origins of heterogeneity. Was there any heterogeneity? The authors need to explain in more detail.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

1. The main problem I see with this work is a possible study selection bias, due to the fact that the authors only considered studies which reported data on HS separately. This likely lead to the exclusion of relevant publications on overall strokes, thus undermining the analysis and consequent comparisons between different stroke types. Therefore, either a different search strategy is used, or results on overall strokes should be eliminated.

Response: Thank you for the comment. It is really true as the reviewer said, the bias undermined the analysis and consequent comparisons between different stroke types. Therefore, we have used different search strategies and have searched more databases to weaken the bias. However, we think there is still a tiny bias existing via our rigorous analysis. To make readers acquire objective results, we have discussed this problem in our discussion section.

2. The discussion on differences between HS overall stroke could be made using data from published reviews as a comparison.

Response: Thank you for the suggestion. In the discussion section, we have used the review from WSO (World Stroke Organization) to discuss the differences between HS and overall stroke.

3. Another important issue concerns criteria for study selection. At the end of the rationale it is stated that retrospective observational studies were reviewed. However, in the Methods section, study design is not reported as a study selection criteria. Actually, Table 1 on study characteristics does feature 1 study, by Sacco et al, defined as prospective and retrospective.

Response: Thank you for the comment. We have rechecked our manuscript. We ensure that retrospective observational studies and prospective and retrospective studies were reviewed in our research. We apologize for our unclear expression. We have corrected the Methods section.

4. TITLE: it should clearly indicate that hospitalizations are investigated

Response: Thank you for your suggestion. Our original title has been revised as "The impact of COVID-19 Pandemic on Hemorrhagic Stroke Admissions: A Systematic Review and Meta-analysis".

5. ABSTRACT: the objective is not clearly stated: again, hospitalizations are the object of investigation.
Response: Thanks for your carefulness. We have corrected this problem and have revised the abstract section.
6. INTRODUCTION: as required in the PRISMA guidelines, the rationale for the review should be described in the context of what is already known (e.g. cite existing reviews or indicate that none is present), and specify the gap that this work intends to fill.
Response: Thank you for the comment. We agree with your suggestion which can benefit our manuscript. We have rewritten the introduction section.
7. METHODS: the search strategy appears very generic and not accurately described, thus it is not replicable. I suggest providing the full strategy as a supplementary file, as indicated by the PRISMA guidelines.
Response: Thank you for the comment. We have provided the full strategy as a supplementary file in our manuscript.
8. RESULTS AND DISCUSSION: please see my introductory comments on stroke types and study selection bias. Also, at the beginning of the discussion, it should be clarified that the focus of the study is admissions, and not patients.
Response: Thanks for your comment. we have revised the result and discussion section.

Reviewer: 2
1. Abstract: Methods: The authors need to specify the statistical analysis methods used in the manuscript.
Response: Thanks for your suggestion. We have added the statistical analysis methods to the abstract section.
2. Methods: The authors used EMBASE, PubMed, and Web of Science electronic databases. Why authors didn't use other sources such as MEDLINE—In Process, Elsevier, Google Scholar, and Cochrane Database of Systematic Reviews (CDSR).
Response: We sincerely thank you for the question, because it gave us a good idea. We have identified studies again through systematically searching EMBASE, PubMed, Web of Science, Elsevier, Medline, Cochrane Library and Google scholar. Seven new articles are added to our quantitative synthesis, which improves the persuasion and reliability of our results.
3. Page 7 line 12: "A fixed-effect model was used when statistically indicated heterogeneity was found (P>.1, I2<50%)" Double check this line I believe you mean A fixed-effect model was used where there was no heterogeneity (P>.1, I2<50%). The authors wrote that Sensitivity analysis was used to exclude the origins of heterogeneity. Was there any heterogeneity? The authors need to explain in more detail.
Response: Thanks for your carefulness. We have corrected this mistake. And there is no heterogeneity through recalculating new data. We actually used sensitivity analysis to assess the impact of individual articles. So we have replaced "Sensitivity analysis was used to exclude the origins of heterogeneity" with "Sensitivity analysis was used to assess the impact of individual articles".

VERSION 2 – REVIEW

REVIEWER	Caminiti, Caterina University Hospital of Parma, Research and Innovation Unit
REVIEW RETURNED	16-Sep-2021

GENERAL COMMENTS	I believe the changes to the manuscript are satisfactory.